

Two Denominators for Turnout

One numerator, two denominators, and a thirty-year decline that is not in the data

84-355 Democracy's Data

August 2026

156,766,239 ballots were counted in the November 2024 general election. Nobody disputes that number. It is the least controversial quantity in this chapter, and it will not change again.

Divide it by one population and American turnout in 2024 was **59.2%**. Divide it by another and it was **64.3%**. The two populations are both published by serious people, both called something reasonable, and both describe the United States in November 2024. The gap between the two answers is 5.1 points, which is 13,500,287 ballots – more votes than were cast in the twelve smallest states combined.

This chapter is about the bottom of that fraction. It is the least examined half of almost every rate in political science. And in this case the choice of denominator, meaning which bottom number you divide by, does not merely move a number. It reverses the answer to the question the number was built to settle.

01 A decline everybody has heard about

By the late 1990s the decline of American voter participation was, in the words of the standard text on the subject, the most important and most analyzed trend in recent American political history. Turnout had run near 62.8% in 1960 and was down to 48.9% by 1996, the lowest figure since the war. An enormous literature grew up to explain it. Television had done it, or negative advertising, or the collapse of party organizations, or the decline of social capital, or the sheer number of elections Americans are asked to vote in.

Every one of those explanations was an explanation of the same measurement: votes cast, divided by the **voting-age population** – everyone in the country aged 18 and over, as estimated by the Census Bureau.

In December 2001 Michael McDonald and Samuel Popkin published a paper in the *American Political Science Review* pointing out that this denominator counts people who are not allowed to vote and misses people who are. The size of the error had been growing since about 1972. That is precisely the period the decline was measured over.¹

The rest of this chapter is what happens when you take their table apart and put it back together.

¹Michael P. McDonald and Samuel L. Popkin, "The Myth of the Vanishing Voter," *American Political Science Review* 95, no. 4 (December 2001): 963–974.

02 A table that prints its own ingredients

There is no dataset behind the 1948–2000 series. There is a table, on page 966 of the December 2001 issue, and it is kept here exactly as a PDF-to-text converter renders it – no character added, removed or straightened.

TABLE1. National Turnout Rates

	Vote for Highest Office	Voting- Age Population	Turnout Rate		Ineligible Adj. Felons	Oversea Adj. VEP	
1948	48833	95573	51.1	2198	+1.2	348	+0.2
1960	68838	109672	62.8	2193	+1.3	481	+0.3
1972	77719	140777	55.2	3640	+1.5	443	+0.2
1984	92653	173995	53.3	13252	+4.4	1153	+0.4
2000	105326	205813	51.2	16500	+4.5	2851	+0.7

This is the whole of the historical evidence, and it is worth pausing on what kind of object it is. It is not a file anybody maintains. It is typesetting. The numbers exist in this arrangement of characters and nowhere else. Every figure in the first half of this chapter was recovered by reading them off the page, the way you would read them off the page.

Public is not the same as usable. This recurs across almost everything this course touches, and it is the reason several chapters begin with a converter rather than a download. A number can be authoritative, cited for twenty-five years, and still not be a dataset.

The parse itself is nearly trivial: 27 lines begin with an election year, and the fields are separated by whitespace. What is not trivial is knowing whether it worked. Splitting on whitespace is exactly the kind of operation that fails silently – read one field too far left and you get numbers that are the right *shape*, in the right columns, and wrong.

03 Rebuilding the paper's own number

The voting-eligible population is not a mysterious quantity. It is the voting-age population, minus the people in it who cannot vote, plus the people who can vote and are not in it:

Term	Who that is
Voting-age population	everyone aged 18 and over living in the country
less noncitizens	in the country, cannot vote
less ineligible felons	citizens, barred by their state over a conviction
plus eligible overseas	citizens who may vote and do not live here
Voting-eligible population	the people who could have cast a ballot

All four of those columns are printed in the table. So the voting-eligible population can be rebuilt, the turnout rate recomputed from it, and the result compared against the rate the authors printed one column further right.

Check	Result
The test	(vote for highest office) / (VAP - noncitizens - felons + overseas)
Election years where the rebuilt rate matches the printed one	27 of 27
Largest disagreement anywhere, percentage points	0.05
Second test	VAP rate + each printed adjustment = VEP rate
Years where the printed adjustments sum to the printed rate	25 of 27

All 27 years reconstruct, the worst of them off by 0.05 of a percentage point, which is rounding in a table printed to one decimal. The second test uses different columns to establish the same thing and agrees.

That is the standard worth holding onto: **a total that adds up is weak evidence that a parse worked.** Two independent properties of the published table, checked against each other, is strong evidence. And a column read one field off would not miss by hundredths – it would miss by tens.

04 The same votes, divided twice

The claim in the literature was a claim about a trend: turnout had been falling since 1972. So take the presidential elections from 1972 to 2000 – 8 of them – and fit a line. Same elections, and the same votes in the numerator, which is the top number. One line for each denominator underneath.

Divided by the voting-age population, presidential turnout fell **0.56 points per election** across 1972–2000. Divided by the voting-eligible population – the same elections, the same ballots, a denominator smaller by a few per cent – **how much did it fall?** Write down a number before you read on. Half as fast? A third?

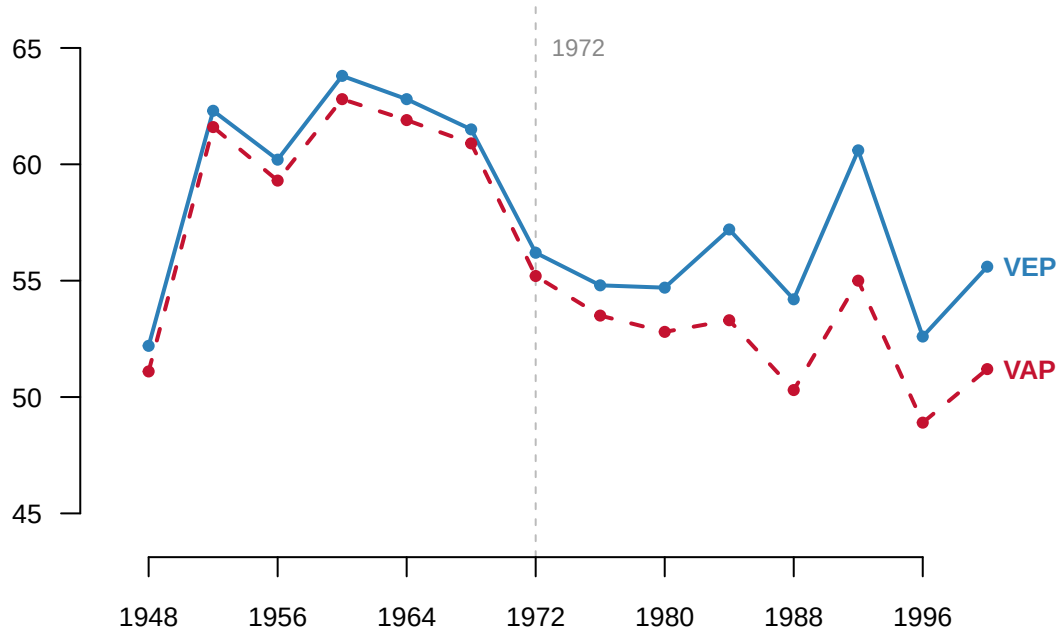
Denominator	1972	2000	Trend per election	Standard error
Voting-age population (VAP)	55.2	51.2	-0.564	0.29
Voting-eligible population (VEP)	56.2	55.6	-0.006	0.40

-0.006 points per election. Not half, and not a third. It is smaller by a factor of about 95, with a standard error of 0.40, sixty times the estimate itself. Twenty-eight years produced an entire research literature on the decline of American civic life. Across them the corrected series falls by 0.04 of a percentage point in total, and cannot be told apart from a flat line.

The published paper reports this coefficient as a drop of 0.007 points per election with a standard error of 0.40. Recomputed here from the printed table, it is -0.006 with a standard error of 0.40 – the same number to the precision the article printed.

Presidential turnout, 1948–2000: the same ballots, two denominators

After 1972 the two lines stop agreeing about the direction of travel



Before 1972 the two lines are nearly the same line. After it they separate, and they separate in a way that matters: the red line goes down and the blue one does not.

05 The error is not a constant

If the voting-age population had always overstated the electorate by the same proportion, none of this would matter for a *trend*. Every year's rate would be too low by the same amount, the line would sit lower, and its slope would be untouched. Anyone studying change over time could ignore the problem entirely.

That is not what happened.

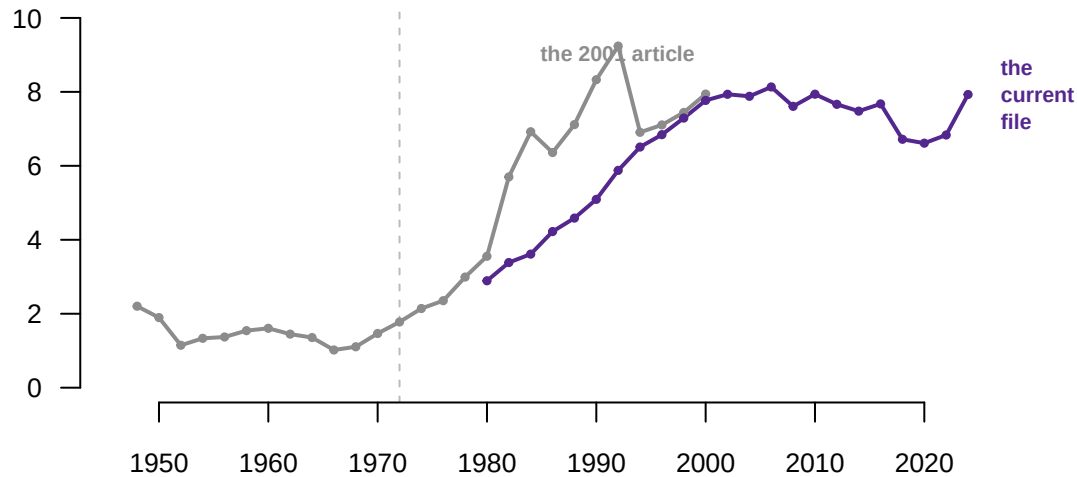
Period	Mean gap between the two rates
1948-1970	0.76 pts
1972-2000	2.78 pts
2024	5.10 pts

The correction is 3.67 times larger in the second period than the first, and by 2024 it is larger again. A growing correction subtracted from a flat series is a declining series. The vanishing voter was never a fact about voters. It was a fact about immigration, incarceration and Census arithmetic, entering the literature through the bottom of a fraction that nobody was reading.

The cleanest way to see this is to drop the numerator entirely. The share of the voting-age population that is not eligible to vote involves no ballots at all – it is a statement about who lives here.

Share of the voting-age population that cannot vote, 1948–2024

No ballots in this figure at all



Around 1.0% at the low point in the 1960s; 7.9% in 2024. That climb is the entire mechanism. Everything in the turnout-decline literature that was attributed to television, to advertising or to social capital had to compete with this line, and for thirty years nobody put it on the same axis.

06 Twenty-one million people

Step	People
Voting-age population, November 2024	264,798,961
less noncitizens	-21,713,515
less people in prison, on probation or on parole where that bars voting	-2,609,565
plus citizens living overseas who may vote	+3,325,217
Voting-eligible population	243,803,423

20,995,538 people, net, and the composition is lopsided: noncitizens are 89.3% of everything subtracted. This is why the correction tracks immigration so closely, and why it grew when it did.

None of these categories is a judgement this chapter is making about who *should* vote. Each is a description of who **may** vote under law as it stands. Whether the felony exclusions in the third row ought to exist is a live political argument, taken up in the disenfranchisement chapter; the arithmetic here is the same whichever side of it you are on.

07 What the modern file actually contains

McDonald has maintained the series ever since, now at the University of Florida, and it is a proper CSV with documentation and a Creative Commons licence – everything the article was not. 1,196 rows, every state and the nation, 23 elections. The first thing to do with it is check that its own headline column is what it says.

Check	Result
Rows in the file	1,196
Rows where VEP = VAP x (1 - noncitizen%) - felons + overseas	1,196 of 1,196
Largest disagreement on that identity	0.058%
Rows carrying a published turnout rate	987
Of those, rows this chapter reproduces within 0.05 points	987 of 987
Largest disagreement with a published rate	0.03 pts

Everything reconciles. And then there are the columns.

51 rows have a web address where a vote count belongs. The column named VOTE_FOR_HIGHEST_OFFICE holds numbers for most of the series and, in 2022, holds links to state election-night reporting sites instead. A tool that coerces that column to a number without looking turns 51 states into missing values, or worse, into zeros.

9 election years have no published turnout rate at all. The VEP_TURNOUT_RATE and VAP_TURNOUT_RATE columns are empty for 1980, 1982, 1984, 1986, 1988, 1990, 1992, 1994, 1996. Every ingredient needed to compute them is present in those same rows; the rate itself simply is not offered. A reader who filters to rows with a rate has silently discarded the first two decades of the file.

And the numerator changes inside the series. The file carries two of them. VOTE_FOR_HIGHEST_OFFICE counts votes for the top statewide contest; TOTAL_BALLOTS_COUNTED counts every ballot, including those where the voter skipped that contest. The second runs two to three per cent higher, and the two cover different years:

Numerator	What it counts	Years available nationally
VOTE_FOR_HIGHEST_OFFICE	votes for the top statewide office	1980-2018 (20 elections)
TOTAL_BALLOTS_COUNTED	every ballot, including undervotes	1998-2024 (14 elections)

Neither covers the whole period. Joining them end to end to get one long series inserts a jump of two to three points at the seam. That jump is larger than most of the effects in this chapter, and it points the same way as the story you were hoping to tell. **Every trend below is computed on one numerator at a time**, which is why the state analysis stops at 2016 rather than running to 2024. That is a real cost, paid deliberately.

08 The same elections, twenty-five years apart

The overlap is a free experiment, and it settles which half of the fraction is the unstable one.

Quantity	Mean change since 2001	Largest single change
Numerator: votes for the highest office	0.004%	0.047%
Denominator: voting-age population	0.783%	2.34%
Result: the published VAP turnout rate	0.35 pts	1.17 pts

The votes have not moved. They were certified before the article went to press and they are the same votes today, to four decimal places.

The population has. The voting-age population of the United States in 2000 was 205,813,000 when the article was published and is 210,623,408 now. That is 4,810,408 people added to a denominator after the fact, because the 2000 Census came in and the estimates that preceded it were revised to meet it. The published turnout rate for that election moved 1.17 points as a result.

A turnout rate has an unstable half and a stable half, and they are not the ones people assume. Election results are contested, litigated, recounted and then fixed forever. Population denominators are uncontroversial, unlitigated, and revised every year for a decade.

09 Fifty states, and five that change direction

Noncitizens are not distributed evenly. In 2024 they are 15.0% of the voting-age population in California and 1.1% in West Virginia. A correction that varies by a factor of 14 across states cannot be waved away as a level shift.

Fit the same two lines for each state across the 10 presidential elections from 1980 to 2016, on one numerator throughout. **5 states come out with turnout falling on one denominator and rising on the other.**

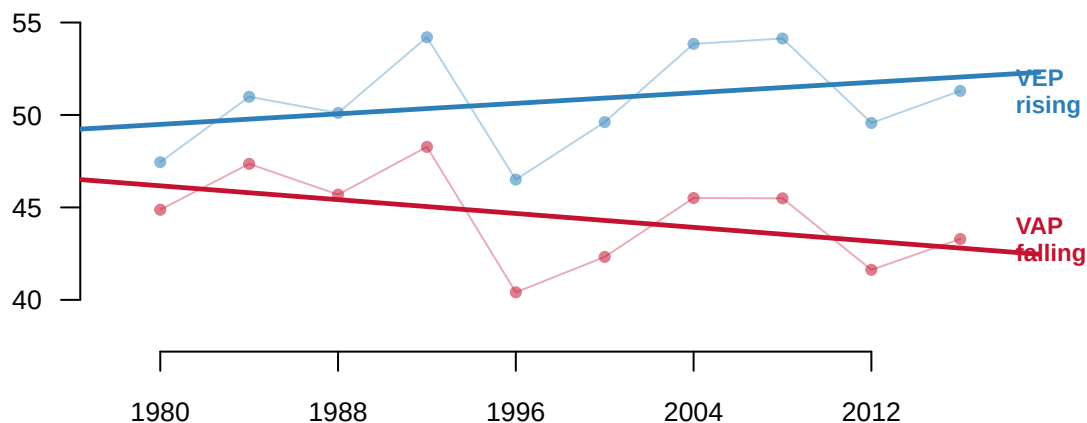
State	Noncitizen share	Fitted change on VAP	Fitted change on VEP
Texas	4.3% to 13.3%	-3.4 pts	+2.6 pts
Arkansas	0.5% to 3.8%	-2.7 pts	+0.3 pts
Kansas	1.1% to 5.2%	-2.5 pts	+0.8 pts
California	10.8% to 16.4%	-2.0 pts	+1.3 pts
Illinois	4.1% to 8.5%	-1.6 pts	+2.0 pts

Texas is the clearest case. Across ten presidential elections its turnout falls 3.4 points measured against everyone of voting age, and rises 2.6 points measured against everyone allowed to vote. That is a swing of 5.9 points, produced by nothing except the choice

of who to count. Over the same period the noncitizen share of its voting-age population went from 4.3% to 13.3%.

Texas presidential turnout, 1980–2016

Falling and rising, at the same time, from the same ballots



Now the honest part, and it is the more useful lesson of the two.

Not one of the 10 trends in that table is statistically distinguishable from zero. 36 of the 102 state trends across all 51 jurisdictions clear the conventional threshold, so this is not a claim that ten elections can never establish anything. It is a claim about these five. The smallest p -value in the flip table is 0.19, and 0 of its 10 slopes are significant.

The sign flips where neither slope is well determined, which is exactly where you would expect it. A trend that is firmly negative stays negative when the denominator shifts a little; a trend near zero is free to land on either side. So the claim that Texas turnout rose is not supported – and neither is the claim that it fell.

That is the finding, not a caveat on it. **The denominator decides which of two unsupported stories the data appears to tell,** and a reader shown only the red line would have had no way to know the blue one existed. Choosing a denominator first and reading the trend off it afterwards is how you get a literature.

10 Two states, and which one voted more

Rankings are how turnout usually reaches the public: best and worst states, league tables, a map coloured by who showed up. Take the 1,275 distinct pairs of states in 2024 and ask how many rank in one order under one denominator and the opposite order under the other.

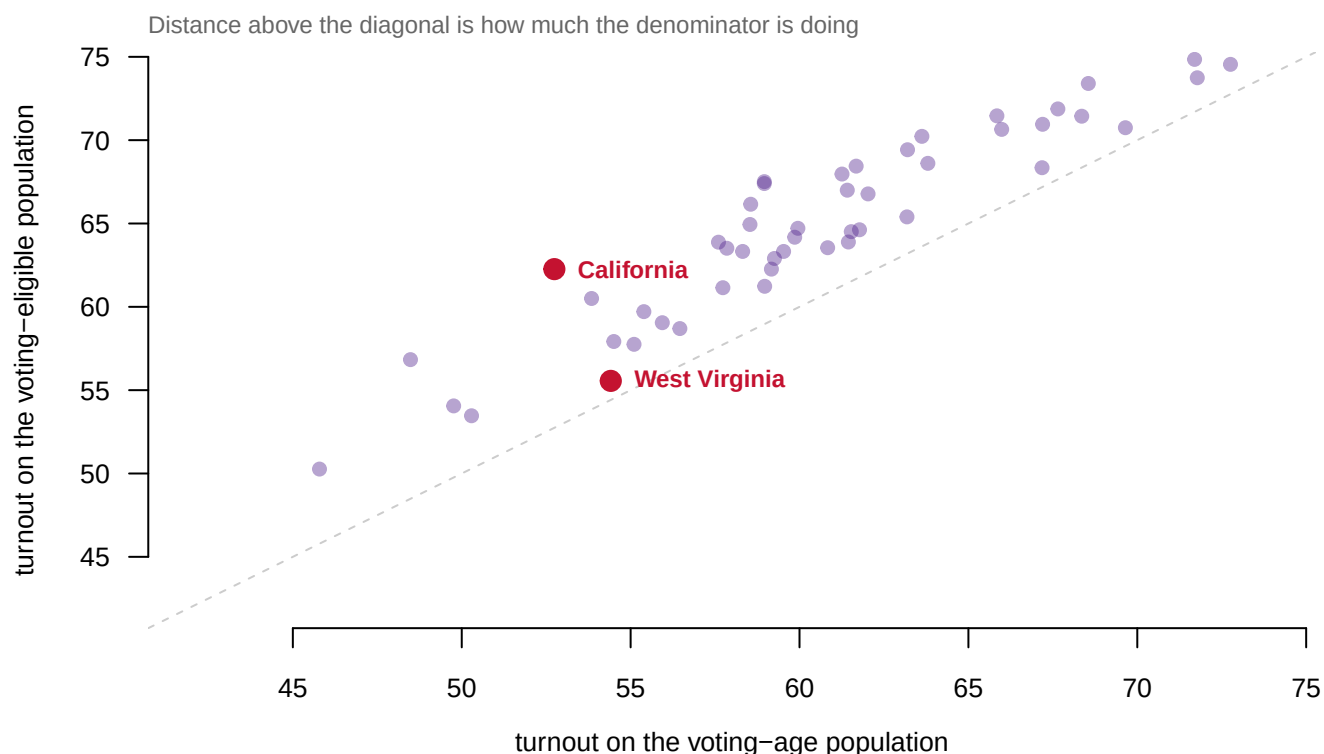
127 of 1,275 – roughly one pair in ten. The largest reversal is not subtle.

Denominator	California	West Virginia	Turned out more
Voting-age population	52.7%	54.4%	West Virginia
Voting-eligible population	62.3%	55.6%	California

West Virginia beats California by 1.7 points, or loses to it by 6.7 points, depending on

which population you put underneath. Both statements are computed from the same certified ballot counts. Unlike the state trends, this one carries no uncertainty worth speaking of: these are two divisions, not two estimates of a slope.

Every state in 2024, measured both ways



Every state sits above the diagonal, because every state has somebody in it who cannot vote. What varies is how far above – from 1.1 points in Vermont to 9.5 points in California. A ranking is a comparison of vertical positions, and the vertical positions have been shifted by different amounts.

11 The better denominator is still wrong

The voting-eligible population subtracts people in prison, on probation and on parole where state law bars them from voting. It does not subtract people who have **completed** a sentence in full and remain barred, and the documentation says so in as many words. The author reports having no reliable way to estimate that population, and therefore chooses not to make the adjustment.²

The disenfranchisement chapter has the number he could not use. The Sentencing Project's *Locked Out 2024* counts 1,616,437 people in that category – Americans who have served their sentences in full, cannot vote, and are inside the denominator this chapter has spent nine pages recommending.

² Michael P. McDonald, 2024 *General Election Turnout Rates*, version 0.4, documentation, United States Elections Project.

Population	Figure
Voting-eligible population, 2024	243,803,423
of which: post-sentence and still barred from voting	1,616,437

Population	Figure
share of the VEP that cannot actually vote	0.66%

That is 0.66% – an order of magnitude smaller than the 7.9% error it replaces, and in the same direction. **The lesson is not that one denominator is right and one is wrong. It is that a denominator is a claim about who counts, that every such claim is contestable, and that the honest move is to say which one you used and why.**

Where this chapter borrows. The post-sentence figure is read straight out of the disenfranchisement chapter's own data folder rather than re-derived here, because that chapter has already made the hard decisions about what the Sentencing Project's five categories mean and which of them a state's law actually bars. Its build script parses them out of the published PDF; this one reads the result.

12 What this evidence will bear

The 1948–2000 series is a table in a journal article, recovered from the printed page and checked two independent ways against properties the table itself publishes. The 1980–2024 series is a maintained CSV under a Creative Commons licence whose every published rate this chapter reproduces to within 0.03 of a point. Both are recomputed each time this document is rendered.

The limits bound different claims, and they are worth taking one at a time.

The denominator is an estimate, not a count. Voting-age population comes from Census projections between decennial counts, and the noncitizen share from the American Community Survey. The felon population comes from Bureau of Justice Statistics series, and the overseas figure from an extrapolation of a 2022 survey. None of them carries a margin of error into this file, meaning any statement of how far off it could be.

The section on the 1980–2000 overlap shows how far they move: up to 2.34% on a single election's voting-age population. **The VEP is a better denominator, not a measured one.**

None of the five sign changes rests on a significant trend. They are real arithmetic on real data, and they are also, every one of them, a change between two slopes that cannot be told apart from zero – which is why they flip at all. They demonstrate how sensitive a conclusion is to a choice of denominator. They do not establish that turnout rose anywhere. The 36 significant trends elsewhere in the same table keep their sign under both denominators, which is the reassuring half of the same result.

The state analysis stops in 2016 because that is where a consistent numerator runs out, and the alternative – splicing two vote-counting definitions at 1998 – would have inserted an artifact larger than the effect. **The 2024 figures are version 0.x** and preliminary; some state ballot counts in that file are visibly rounded estimates rather than certifications.

Two elections carry no highest-office numerator at all, so 2020 and 2024 appear only in total-ballots form.

And the better denominator still contains 1,616,437 people who cannot vote, by the maintainer's own documented choice, plus everyone excluded by residency requirements, which neither source adjusts for.

What can be said is bounded and worth saying. The decline in American voter turnout after 1972 is the most analyzed trend in modern American political history, and the subject of an enormous literature on television, advertising and social capital. Divide the ballots by the people who were allowed to cast them, and it is -0.006 points per election, with a standard error of 0.40. The correction that produces this is not exotic and was never secret. It is subtraction, applied to a column that everybody had. Nobody applied it for thirty years because the denominator is the half of a rate that nobody reads.

13 Sources

Data

- McDonald, Michael P., and Samuel L. Popkin. 2001. "The Myth of the Vanishing Voter." *American Political Science Review* 95(4): 963-974. <https://www.jstor.org/stable/3117725> Table 1, page 966, captured with `pdftotext -layout` into `mp2001-table1.txt`. 27 election years, 1948-2000. See `README.txt` for what was and was not altered.
- McDonald, Michael P. *1980-2022 General Election Turnout Rates*, version 1.2. United States Elections Project, University of Florida. CC BY 4.0. https://election.lab.ufl.edu/data-downloads/turnoutdata/Turnout_1980_2022_v1.2.csv
- McDonald, Michael P. *2024 General Election Turnout Rates*, version 0.4. United States Elections Project, University of Florida. https://election.lab.ufl.edu/data-downloads/turnoutdata/Turnout_2024G_v0.4.csv Both files' documentation is fetched alongside them and quoted above.
- Uggen, Christopher, Ryan Larson, Sarah Shannon and Robert Stewart. 2024. *Locked Out 2024: Four Million Denied Voting Rights Due to a Felony Conviction*. The Sentencing Project. Read from `national.csv`, which that chapter's build script parses out of the published PDF.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`mp1948.csv`, `national.csv`, `overlap.csv`, `pairs2024.csv`, `state_trends.csv`, `states.csv`, `national.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `state_trends.csv` fits a trend on both denominators and flags `sign_flip`. Count the states where the two denominators disagree about the direction of turnout.
- `pairs2024.csv` compares states in pairs on both denominators, with a `swing`. Find the pairs where the choice of denominator reverses which state turned out more.
- `mp1948.csv` shows the adjustments one at a time: noncitizens, felons, overseas. Work out which adjustment matters most, and whether that has changed over time.
- `overlap.csv` recomputes an old published series with today's numbers. Compare `vap_rate_move` and `vap_rate_move`. How much of a famous decline survives?

Also cited

- Rosenstone, Steven J., and John Mark Hansen. 1993. *Mobilization, Participation, and Democracy in America*. New York: Macmillan — the characterisation of the decline as the most analyzed trend in recent American political history, quoted at the opening of the 2001 article.
- Teixeira, Ruy. 1992. *The Disappearing American Voter*. Washington, DC: Brookings Institution — the case for the voting-age population as the denominator, which the 2001 article argues against.
- Putnam, Robert D. 2000. *Bowling Alone*. New York: Simon and Schuster — one of the explanations offered for the decline that turns out not to need explaining.

Checks

Every figure below is recomputed from the sources listed above, and checked against the values asserted in `checks.csv`. The table is printed here verbatim.

Check	Value
Table 1 data rows parsed	27
Table 1 election years	1948-2000
rows where the rebuilt VEP rate matches the printed one within 0.1 pts	27
largest disagreement between rebuilt and printed VEP rate, pts	0.05
rows where the printed adjustments sum to the printed VEP rate within 0.2 pts	25
paper's VEP trend 1972-2000, pts per election	-0.006
its standard error	0.40
paper's VAP trend over the same years, pts per election	-0.564
Elections Project rows, 1980-2024	1,196
rows where VEP equals VAP*(1-noncitizen%) - felons + overseas within 0.1%	1,196
largest disagreement on that identity, %	0.058
rows with a published turnout rate to check against	987
of those, rows reproduced within 0.05 pts	987
largest disagreement with a published rate, pts	0.03
rows where VOTE_FOR_HIGHEST_OFFICE holds a URL	51
which years those are	2022
election years with no published turnout rate at all	1980, 1982, 1984, 1986, 1988, 1990, 1992, 1994, 1996
2024 national VAP-to-VEP gap, pts	5.10
ballots that gap is worth	13,500,186
state pairs compared in 2024	1,275
pairs that reverse under the two denominators	127
states whose 1980-2016 presidential trend changes sign	5
elections published in both 1980-2000 sources	11
mean revision to the numerator since 2001, %	0.004
mean revision to the denominator since 2001, %	0.783

Check	Value
largest revision to a published VAP turnout rate, pts	1.17

Every file this chapter reads can be regenerated from the addresses above. For each download the url, bytes, hash and row count are recorded and compared against the previous run, so a source that changes under the chapter announces itself instead of passing quietly.

REBUILD THIS DATA YOURSELF, WITH AN AI ASSISTANT

I want to rebuild a public dataset from its original sources, without any starting files. Please do the following and show your work.

THE FIRST SOURCE. The United States Elections Project at the University of Florida, general-election turnout rates, two CSVs:

https://election.lab.ufl.edu/data-downloads/turnoutdata/Turnout_1980_2022_v1.2.csv

https://election.lab.ufl.edu/data-downloads/turnoutdata/Turnout_2024G_v0.4.csv

Each row is one state (or the national total) in one election, with the voting-age population, the voting-eligible population, and the pieces separating them. No account or key is needed. The version numbers in those addresses will move – if one answers 404, start from <https://election.lab.ufl.edu/data-downloads/> and take the current version. A 0.x version means the publisher still calls it preliminary.

THE SECOND SOURCE. McDonald and Popkin, "The Myth of the Vanishing Voter", American Political Science Review 95(4), 2001, Table 1 on page 966 – the 1948–2000 half of the series exists only as a table printed in that article. It sits behind JSTOR (<https://www.jstor.org/stable/3117725>), so it cannot be fetched openly; anyone with library access can read it there and key in its 27 rows by hand.

WHAT TO BUILD.

1. From the article's table: the printed VEP turnout rate for each election, and the same rate rebuilt from the table's own component columns, to confirm the arithmetic of the paper.
2. From the modern files: national turnout on both denominators, 1980–2024 – keeping the two numerators (highest-office votes and

total ballots) apart, because the file's own rate columns switch numerators partway and splicing them fakes a jump around 1998.

CHECK YOUR WORK. The copies this book used were fetched in August 2026. If your rebuild matches them, you should find:

- 27 data rows in the article's table, 1948 through 2000, and the rebuilt VEP rate lands within 0.1 points of the printed one in all 27
- 1,144 rows in the 1980-2022 file and 52 in the 2024 file
- across all 1,196 modern rows, the eligible population equals the voting-age population less noncitizens and felons, plus overseas voters, within a tenth of a percent

The publisher revises these files and bumps the version string; a 2024 file past v0.4 means numbers were revised, not that you erred.